Overview of Ridge regression

Subject: Ridge regression

$$L(\beta) = \|y - X\beta\|^2 + \lambda\|\beta\|^2$$

$$\hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

1.

where we have used ■ x ■ Q 2 {\displaystyle \left\|\mathbf {x} \right\|_{Q}^{2}} to stand for the weighted norm squared x T Q x {\displaystyle \mathbf {x} ^{\mathsf {T}}Q\mathbf {x} } (compare with the Mahalanobis distance). In the Bayesian interpretation P {\displaystyle P} is the inverse covariance matrix of b {\displaystyle \mathbf {b} } , x 0 {\displaystyle \mathbf {x} _{0}} is the expected value of x {\displaystyle \mathbf {x} } , and Q {\displaystyle Q} is the inverse covariance matrix of x {\displaystyle \mathbf {x} } . The Tikhonov matrix is not explicitly included because the corresponding regularization term ■ Γ x − x 0 ′ ■ Q ′ 2 {\displaystyle \left\|\Gamma \mathbf {x} -\mathbf {x} _{0}'\right\|_{Q'}^{2}} reduces to above with Γ x 0 = x 0 ′ {\displaystyle \Gamma \mathbf {x} _{0}=\mathbf {x} _{0}'} and Q = Γ T Q ′ Γ {\displaystyle Q=\Gamma ^{T}Q'\Gamma } . For normal regularization where Q ′ = I {\displaystyle Q'=I} , the Tikhonov matrix then appears in the Cholesky factorization Q = Γ T Γ {\displaystyle Q=\Gamma ^{\mathsf {T}}\Gamma } and is considered a whitening filter. This generalized problem has an optimal solution x ^ {\displaystyle {\hat {\mathbf {x} }}} which can be written explicitly using the formula

2.

The effect of regularization may be varied by the scale of the matrix Γ {\displaystyle \Gamma } . For Γ = 0 {\displaystyle \Gamma =0} this reduces to the unregularized least-squares solution, provided that (ATA)−1 exists. Note that in case of a complex matrix A {\displaystyle A} , as usual the transpose A T {\displaystyle A^{\mathsf {T}}} has to be replaced by the Hermitian transpose A H {\displaystyle A^{\mathsf {H}}} . L2 regularization is used in many contexts aside from linear regression, such as classification with logistic regression or support vector machines, and matrix factorization.

3.

History Tikhonov regularization was invented independently in many different contexts. It became widely known through its application to integral equations in the works of Andrey Tikhonov and David L. Phillips. Some authors use the term Tikhonov–Phillips regularization. The finite-dimensional case was expounded by Arthur E. Hoerl, who took a statistical approach, and by Manus Foster, who interpreted this method as a Wiener–Kolmogorov (Kriging) filter. Following Hoerl, it is known in the statistical literature as ridge regression, named after ridge analysis ("ridge" refers to the path from the constrained maximum).

4.

which shows that λ {\displaystyle \lambda } is nothing but the Lagrange multiplier of the constraint. In fact, there is a one-to-one relationship between c {\displaystyle c} and λ {\displaystyle \lambda } and since, in practice, we do not know c {\displaystyle c} , we define λ {\displaystyle \lambda } heuristically or find it via additional data-fitting strategies, see Determination of the Tikhonov parameter below. Note that as λ ↓ 0 {\displaystyle \lambda \downarrow 0} , the constraint eventually becomes non-binding, and the ridge estimator converges to the minimum-norm ordinary least squares estimator, here denoted as β ^ = β ^ 0 {\displaystyle {\hat {\boldsymbol {\beta }}}={\hat {\boldsymbol {\beta }}}_{0}} :

5.

where Y {\displaystyle \mathbf {Y} } is the regressand or response vector, X {\displaystyle \mathbf {X} } is the design matrix, I {\displaystyle \mathbf {I} } is the identity matrix, and the ridge (or Tikhonov) regularization parameter λ ≥ 0 {\displaystyle \lambda \geq 0} serves as the constant shifting the diagonals of the moment matrix. It can be shown that this estimator is the solution to the least squares problem subject to the constraint β T β = c {\displaystyle {\boldsymbol {\beta }}^{\mathsf {T}}{\boldsymbol {\beta }}=c} , which can be expressed as a Lagrangian minimization:

6.

where the Wiener weights are f i = σ i 2 σ i 2 + α 2 {\displaystyle f_{i}={\frac {\sigma _{i}^{2}}{\sigma _{i}^{2}+\alpha ^{2}}}} and q {\displaystyle q} is the rank of A {\displaystyle A} .

7.

Application to existing fit results Since Tikhonov Regularization simply adds a quadratic term to the objective function in optimization problems, it is possible to do so after the unregularised optimisation has taken place. E.g., if the above problem with Γ = 0 {\displaystyle \Gamma =0} yields the solution x ^ 0 {\displaystyle {\hat {\mathbf {x} }}_{0}} , the solution in the presence of Γ ≠ 0 {\displaystyle \Gamma \neq 0} can be expressed as:

8.

where A = (A Γ) {\displaystyle {\mathcal {A}}={\begin{pmatrix}A\\\Gamma \end{pmatrix}}} and b = (b 0) {\displaystyle {\mathcal {b}}={\begin{pmatrix}\mathbf {b} \\{\boldsymbol {0}}\end{pmatrix}}} , for some suitably chosen Tikhonov matrix Γ {\displaystyle \Gamma } . In many cases, this matrix is chosen as a scalar multiple of the identity matrix (Γ = α I {\displaystyle \Gamma =\alpha I}), giving preference to solutions with smaller norms; this is known as L2 regularization. In other cases, high-pass operators (e.g., a difference operator or a weighted Fourier operator) may be used to enforce smoothness if the underlying vector is believed to be mostly continuous. This regularization improves the conditioning of the problem, thus enabling a direct numerical solution. Treating it as an ordinary least squares problem with augmented matrices A {\displaystyle {\mathcal {A}}} and b {\displaystyle {\mathcal {b}}} , the solution is

9.

with the "regularisation matrix" B = (A T A + Γ T Γ) − 1 A T A {\displaystyle B=\left(A^{\mathsf {T}}A+\Gamma ^{\mathsf {T}}\Gamma \right)^{-1}A^{\mathsf {T}}A} . If the parameter fit comes with a covariance matrix of the estimated parameter uncertainties V 0 {\displaystyle V_{0}} , then the regularisation matrix will be

10.

λ ∗ := ς 2 t r (Ω) β ■ Ω β + 3 ς 2 t r (Ω 2) / n , {\displaystyle \lambda ^{*}:={\frac {\varsigma ^{2}\mathrm {tr} (\Omega)}{{\boldsymbol {\beta }}^{\top }\Omega {\boldsymbol {\beta }}+3\varsigma ^{2}\mathrm {tr} (\Omega ^{2})/n}},}

11.

Determination of the Tikhonov parameter The optimal regularization parameter λ {\displaystyle \lambda } is usually unknown and in practice needs to be estimated. Typically, a data-driven choice for the Tikhonov regularization parameter λ {\displaystyle \lambda }

12.

Tikhonov regularization for linear equations Suppose that for a known real matrix A {\displaystyle A} and vector b {\displaystyle \mathbf {b} } , we wish to find a vector x {\displaystyle \mathbf {x} } such that